FinServ Global Regulatory Compliance Assistant

Solution Architecture Document
Dr. Ramkumar Gurusamy, AI Architect technical assignment, November 2026.

1. Summary

The problem. FinServ Global has 40 compliance officers. They spend more than 60% of their time searching regulatory PDFs and preparing evidence. More than 200 new circulars and amendments arrive each year across Basel III, MiFID II and RBI directions. One cross-border trade can fall under several frameworks, and regulators expect a documented reason for every decision.

The approach. Compliance is a traceability problem first and a language problem second. The design follows five rules:

1. **Rules decide, the LLM explains.** Risk ratings come from versioned rules that people can review. The LLM writes the explanation but cannot change a rating.
2. **Regulation is versioned by paragraph and date.** Any answer can be reproduced as of the day it was given.
3. **Every answer is checked before it is shown.** If a citation is invalid or a number is not in the source, the system retries once, then shows the quoted source text instead.
4. **Data residency is enforced in code.** The LLM gateway checks where the data may go before every call. If no model is allowed, the call is blocked.
5. **Failures are visible.** If the models are down, users get quoted sources marked as degraded. They never get an answer without citations.

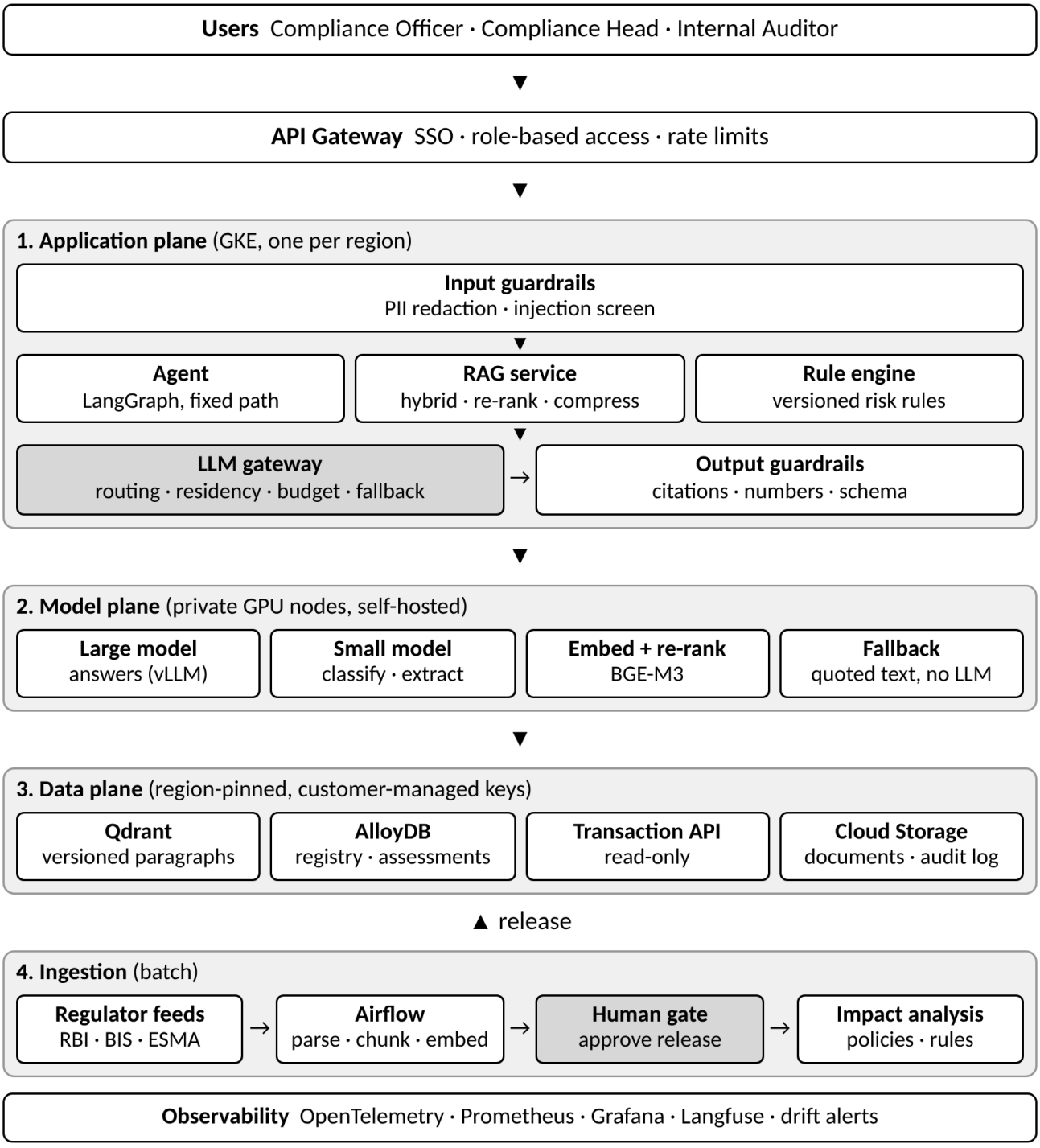
The prototype is a working slice of this design. Section 10 lists what is built and tested and what is design only.

2. Requirements

Area	Requirement	Target
Users	Officers, Compliance Head, Internal Auditor	Single sign-on with role-based views, in three regions
Load	500 concurrent users, 10,000 queries a day	Peak of about 6.7 requests a second (see 8.3)
Q&A speed	Fast, cited answers	Full answer under 8 s (p95), first words under 1.5 s
Screening speed	Interactive	Under 15 s (p95)
Availability	Business critical, but not in the payment path	99.9% per region; lose at most 15 min of data; recover within 1 hour
Evidence	Every decision must be traceable	100% of AI-assisted outputs in a tamper-evident log
Freshness	200+ changes a year	New regulation reviewed and live within 1 business day
Quality	No unsupported regulatory claims	Faithfulness at least 0.85; 100% of unanswerable questions refused

To confirm in discovery: legal entity structure, any regulator conditions on cloud outsourcing, retention periods, the system of record for transactions, and which US rules are in scope. The assignment names Basel III, MiFID II and RBI, so US rules are an extension, not part of the current corpus.

3. Architecture overview



The system has four planes. Each can change without affecting the others.

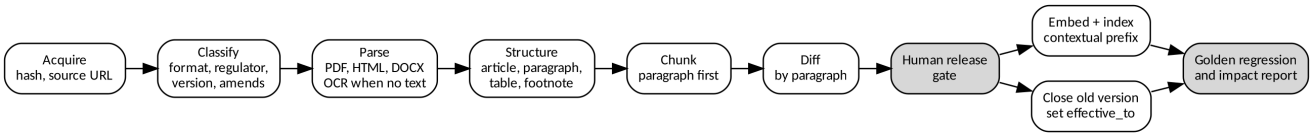
Plane	What it does	Why it is separate
Application	APIs, agent, rules, guardrails	No stored state, runs on CPU, released often

Plane	What it does	Why it is separate
Model	Text generation, embeddings, re-ranking	Runs on GPUs, which cost and scale differently; models change on their own schedule
Data	Vectors, keyword index, registry, audit log	Pinned to one region, with the strictest access controls
Ingestion	Collect, parse, version and release regulation	Batch work with a human approval step, not on the query path

Why one LLM gateway? Every model call goes through it. That gives one place to enforce residency, budgets and fallback, to count tokens, and to swap models without changing application code. The prototype's `ModelRouter` is a working version of the gateway.

4. RAG pipeline

4.1 Ingestion



Format	Tool	How it is handled
Digital PDF	Docling (open source); PyPDF in the prototype	Keeps layout, headings and tables
Scanned PDF	Tesseract; Document AI as a paid option	Low text yield sends it to OCR, with quality sampling
HTML circulars	BeautifulSoup with one extractor per site	RBI and ESMA pages each have their own template
DOCX	python-docx through Docling	Tracked changes are flattened and logged

Metadata matters as much as text. Each chunk stores the regulator, framework, jurisdiction, document reference, version, effective dates, paragraph reference, source URL, content hash and residency class. Citations and the audit log depend on these fields. A document with missing or invalid metadata is held back, not indexed with defaults.

Watching for changes. Scheduled jobs check regulator feeds and pages. A new or changed document opens a review task, and nothing goes live until a compliance reviewer releases it. This can add up to one business day. That is deliberate: a wrong regulation served quickly is worse than a correct one served a day later.

4.2 Chunking

Option	Benefit	Problem	Decision
Fixed 512 tokens	Simple	Cuts clauses and breaks citations	Rejected
Recursive character split	Better boundaries	Does not know paragraph numbers	Rejected
Semantic split	Follows topics	Changes between runs and is hard to cite	Rejected
Paragraph first, sentence windows if too long	The chunk is the unit a regulator cites	Paragraph lengths vary	**Selected**

Rules:

- One numbered paragraph is one chunk: up to 220 tokens in the prototype, 350 in production.
- A longer paragraph is split at sentence boundaries with about 15% overlap. Every piece keeps the same paragraph reference.
- Definitions, tables and footnotes stay with the paragraph they belong to. Tables are also stored as rows, because the answer is often a percentage in a table.
- Each chunk is embedded with a short prefix (framework, paragraph reference, heading), so a clause on its own still shows which regulation it comes from.

Overlap is kept small on purpose. Regulatory paragraphs are written to stand alone, and large overlap would fill the results with near-duplicates, pushing out other frameworks that a cross-regulation question needs.

4.3 Embedding model

Model	Licence	Size	Strength	Weakness	Use
BGE-M3	MIT	1024	Dense and sparse output, long inputs, EU languages	Heavier than small models	**Production**
BGE-small-en-v1.5	MIT	384	Fast on CPU	English only, less precise	Laptop
WordLlama L2	MIT	256	Installed with pip, no download	Lower quality	Offline tests
Vertex AI embeddings	Proprietary	Up to 3072	Managed, high quality	Vendor lock-in, cost per token	Not the default

Self-hosted embeddings also keep query text, which can include client details, inside the region. A finance-specific fine-tune is possible later, but only if evaluation shows it helps, because it means re-embedding the whole corpus.

4.4 Vector database

Qdrant, self-hosted in each region. It has strong metadata filtering, built-in sparse vectors and tenant isolation. pgvector was the closest alternative. See ADR-001.

4.5 Retrieval steps

1. **Understand the query.** A small model labels the intent and framework. A glossary expands terms, for example `CDD` to `customer due diligence`. Expansions are used to find candidates, not to score them, because scoring on them ranked all KYC chunks equally.
2. **Filter.** Date (`as_of`), framework, jurisdiction and user permissions are applied before search.
3. **Hybrid search.** Dense search and BM25 keyword search each return 30 candidates. Keyword search is needed because regulators cite exact tokens such as `RBC20.1`, `Article 25(3)` and `4.5%`.
4. **Merge.** Reciprocal Rank Fusion (RRF) combines the two lists by rank, so scores need no calibration.
5. **Re-rank.** A cross-encoder re-scores the top 20 and keeps 6. Production uses `bge-reranker-v2-m3`. This step stays only while tests show it improves results (see 8.5).
6. **Compress.** Only the most relevant original sentences are kept. Compression selects sentences; it never rewrites legal text.
7. **Refuse when unsure.** If the evidence is weaker than a set threshold, the system says it cannot answer.

4.6 Updates and versioning

- **Nothing is deleted.** Each chunk has `effective_from` and `effective_to` dates. A new version closes the old chunk.
- **Partial amendments stay partial.** A circular that changes paragraph 50 closes only paragraph 50. The rest of the master direction stays live.
- **IDs are stable.** `chunk_id = hash(doc_id, version, paragraph, ordinal)`, so running ingestion twice gives the same result.
- **Point-in-time questions.** Filtering on effective dates answers "what applied on 14 March?", so an auditor can replay an old decision against the rules in force at the time.
- **Safe re-indexing.** A new embedding model builds a new collection, must pass the test set, and then an alias switches over. The old collection stays for rollback.

When sources conflict, both are shown. For example, an RBI rule and the Basel standard are both cited, and the stricter one is flagged for an officer.

5. LLM orchestration

5.1 Model selection

Open models change quickly, so the design fixes **how a model is chosen**. The models below are the current shortlist. The final choice needs a benchmark on FinServ's own test set.

Tier	Task	Shortlist	Hosting	Why
Fast	Classify, extract, route	Gemma 4 small variants, Granite 3.3 2B	vLLM or CPU	Low cost, permissive licences
Main	Cited answers, narratives	Gemma 4 31B or Qwen 3.6 35B	vLLM on one L4 GPU	Apache 2.0; reported to fit 24 GB when quantised
Escalation	Reasoning across frameworks	Mistral Small 4 (needs at least 48 GB, as reported)	vLLM on a larger GPU	Only if tests show the main tier falls short
Optional, **proprietary**	Managed fallback in region	Gemini on Vertex AI, regional endpoint	Google-managed	Only after legal review; the open fallback stays the default

Selection order: licence allowed, can run in region, faithfulness on the test set, p95 latency under load, then cost.

Replacing a model. Weights and tokenisers are pinned by digest in a private registry. A new model first runs in shadow, then takes 5%, 25% and 100% of traffic. If faithfulness drops 0.03 below the current model, or refusal accuracy drops at all, it rolls back automatically.

5.2 Routing

Request	Route
Intent labelling, field extraction	Fast tier
Factual question on one framework	Main tier
Several frameworks, or a screening narrative	Main tier; escalation tier if validation fails
Provider failure, circuit open or residency denied	Next allowed provider; finally the extractive fallback

Request	Route
Daily budget used up	Paid external tiers stop; self-hosted tiers continue

Every call goes through four checks in order: residency, budget, circuit breaker, then the call itself. The prototype implements and tests all four.

5.3 Prompts

Prompts are kept in version control (`config/prompts.yaml`), and each answer records the prompt version used. A prompt change must pass the same evaluation as a code change.

Part	Content
System	Role, answer only from sources, citation format, copy numbers exactly, refusal phrase, no legal rulings
Examples	One cited answer and one refusal, so a small model learns that refusing is allowed
Reasoning	Hidden steps: find sources, check versions, note conflicts, write supported sentences. The steps are not shown, so the output stays clean and unsupported reasoning does not reach users
Task	Numbered sources with paragraph and version
Retry	States exactly which check failed

Prompt injection. Retrieved text is marked as reference data, and the model is told to ignore any instructions inside it. Inputs are screened for known attack patterns. The main protection is that the agent's tools are read-only, so a successful injection has nothing to act on.

5.4 Guardrails

Stage	Check	Method	If it fails	In prototype
Input	PII removal	Regex now; Presidio or GLiNER in production	Redact and count	Yes (regex)
Input	Injection screen	Pattern match	Refuse and log	Yes
Retrieval	Weak evidence	Evidence threshold	Refuse	Yes
Retrieval	Permissions	Metadata filter	Exclude	Design
Output	Valid citations	Each label must match a retrieved chunk	Retry, then fallback	Yes
Output	Correct numbers	Every number must appear in the cited source	Retry, then fallback	Yes
Output	Hallucination	Sentence-level entailment model	Retry, then fallback	Design; simple rule-based check in prototype
Output	Structure	Pydantic schema validation	Reject	Yes
Decision	Human review	Critical or incomplete cases go to an officer	Queue	Yes (status)

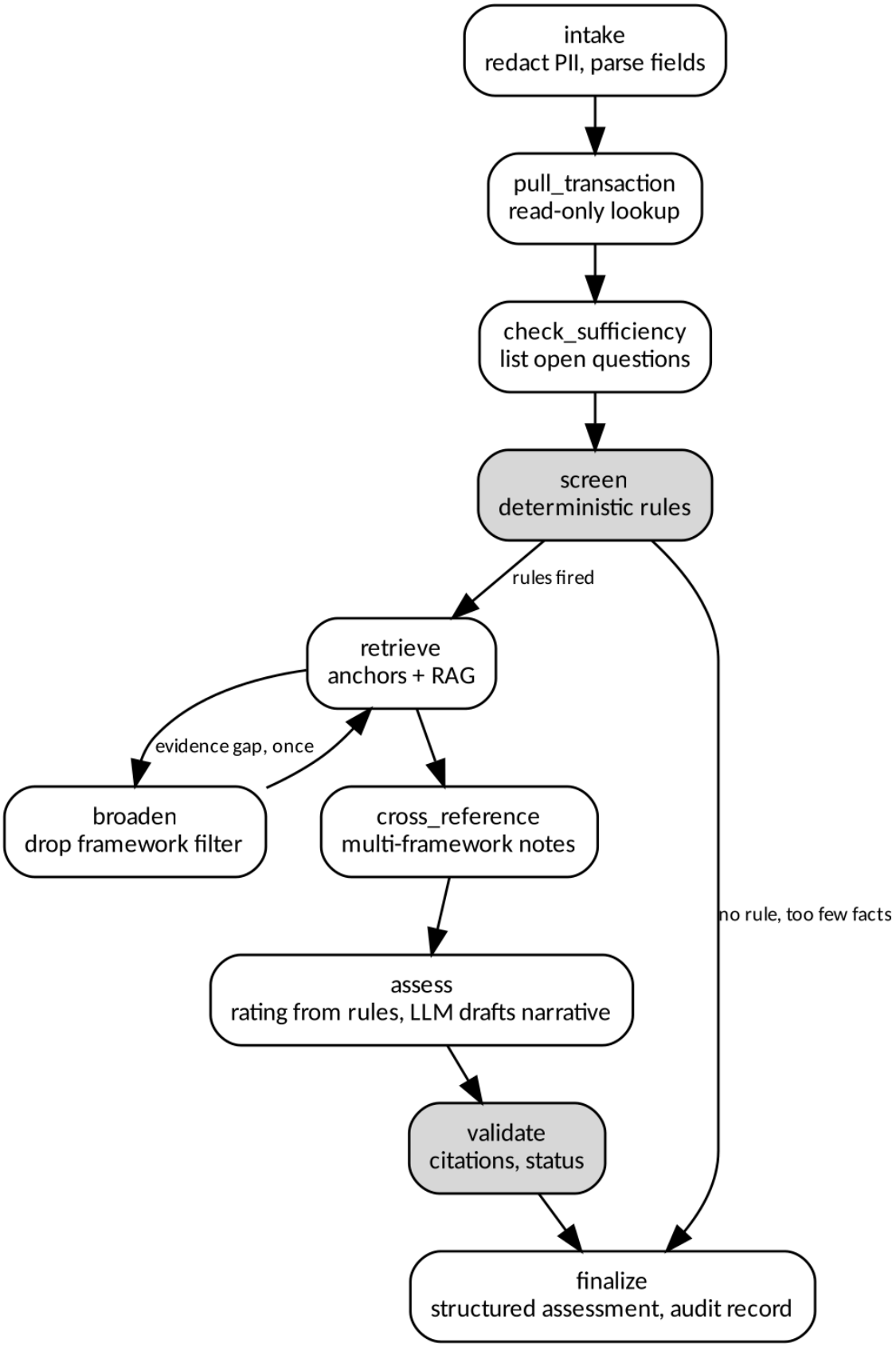
ADR-003 explains why these checks are built into the application instead of relying only on a guardrail framework.

6. Agent design

6.1 Framework

LangGraph, used as a fixed state machine, not a free-running agent. It gives typed state, conditional steps, checkpoints and a trace that can be replayed. CrewAI's role-based conversations are harder to audit, and plain code would mean rebuilding checkpoints and tracing. See ADR-002.

6.2 Workflow



The key choice is **what the LLM is not allowed to do**. It does not choose tools, set a rating or decide which regulation applies. Those steps are fixed code. A free-running agent would be more flexible, but its path could change from run to run, which does not suit a decision that must be audited. Retrieval supplies evidence for the rules, and the LLM only writes the explanation, which must pass validation.

6.3 Tools

Tool	Input	Output	Changes data?	If it fails
`query_regulations`	Query, frameworks, as_of date	Ranked, versioned chunks	No	Retry once with a broader search, then human review
`get_paragraph`	Paragraph reference	Current paragraph	No	Flag an evidence gap
`pull_transaction`	Transaction ID	Transaction record	No (read-only)	Record the error, human review
`screen_rules`	Transaction	Rules triggered	No	Schema error stops the run
`generate_narrative`	Findings, sources	Cited summary	No	Use template text
`write_audit`	Event	Audit ID	Append only	Run stops (fails closed) in production

6.4 State and errors

The agent's state holds: raw input, cleaned transaction, trace, triggered rules, evidence, gaps, findings, questions, status and rating. Production saves checkpoints to AlloyDB, so a run continues after a restart. The prototype keeps checkpoints in memory.

Situation	What happens
Missing facts	Unknown values stay empty and never lower the risk. The system gives a provisional rating and asks specific questions
Too little information	Status `INSUFFICIENT_INFO`, rating `UNDETERMINED`; no guessing
Search fails	One broader retry, then `HUMAN_REVIEW`; the rule-based rating stands
Transaction not found	Recorded; `HUMAN_REVIEW`
LLM fails	Template narrative is used; rating and citations are unaffected
Several frameworks apply	Each obligation is listed separately, and the stricter one is flagged

6.5 Change impact and reports

- **New regulation.** Changed paragraphs are matched against the policy register and the rules that cite them. Direct links are reported as affected. Similar-sounding items are only listed as "possible" for review.
- **Compliance reports.** Reports group assessments by risk, framework and finding. Each line links to the audit log by assessment ID. All figures come from stored data, not from a model.

7. Security and data residency

7.1 Data classes

Class	Examples	Where it can go
Public regulation	Basel text, published circulars	Any approved region

Class	Examples	Where it can go
Internal	Policies, reports, prompts	The entity's own region
Confidential	Transactions, client details	The entity's own region, self-hosted models only
Restricted	ID documents, account numbers	Removed before any model call

Regulatory text is public, so the sensitive part of a prompt is usually the transaction. Even so, RBI-tagged text is treated as India-only by default, because the same prompt often carries client details, and because that is the safe reading of the client's requirement.

7.2 Stopping data leaks to external providers

1. **Self-hosted by default.** All models run in the same region as the data, so no core flow needs an external LLM.
2. **Network.** GKE nodes have no public IP. VPC Service Controls and a default-deny egress policy block model pods from the internet.
3. **Code.** The gateway refuses any provider not cleared for the data's residency class. If none is cleared, the call is blocked. Tests confirm the external endpoint receives no request.
4. **Redaction.** PII is removed before prompts and logs.
5. **Contract.** Any optional proprietary service needs zero data retention, no training on the data, processing in region, and legal sign-off.

7.3 Encryption, access and audit

Control	Design
Data at rest	AES-256 with customer-managed keys in Cloud KMS, one key ring per region
Data in transit	TLS 1.3 at the edge; mutual TLS between services through a service mesh
Identity	OIDC single sign-on; Workload Identity for pods; no long-lived keys
Model access	Only the gateway's service account can call the models
Permissions	Role and entity checks on search filters and APIs
Audit	Each record links to the previous one by hash, and stores the input, retrieved chunks and versions, model, prompt, rule version, guardrail results and output. Records are archived to locked Cloud Storage

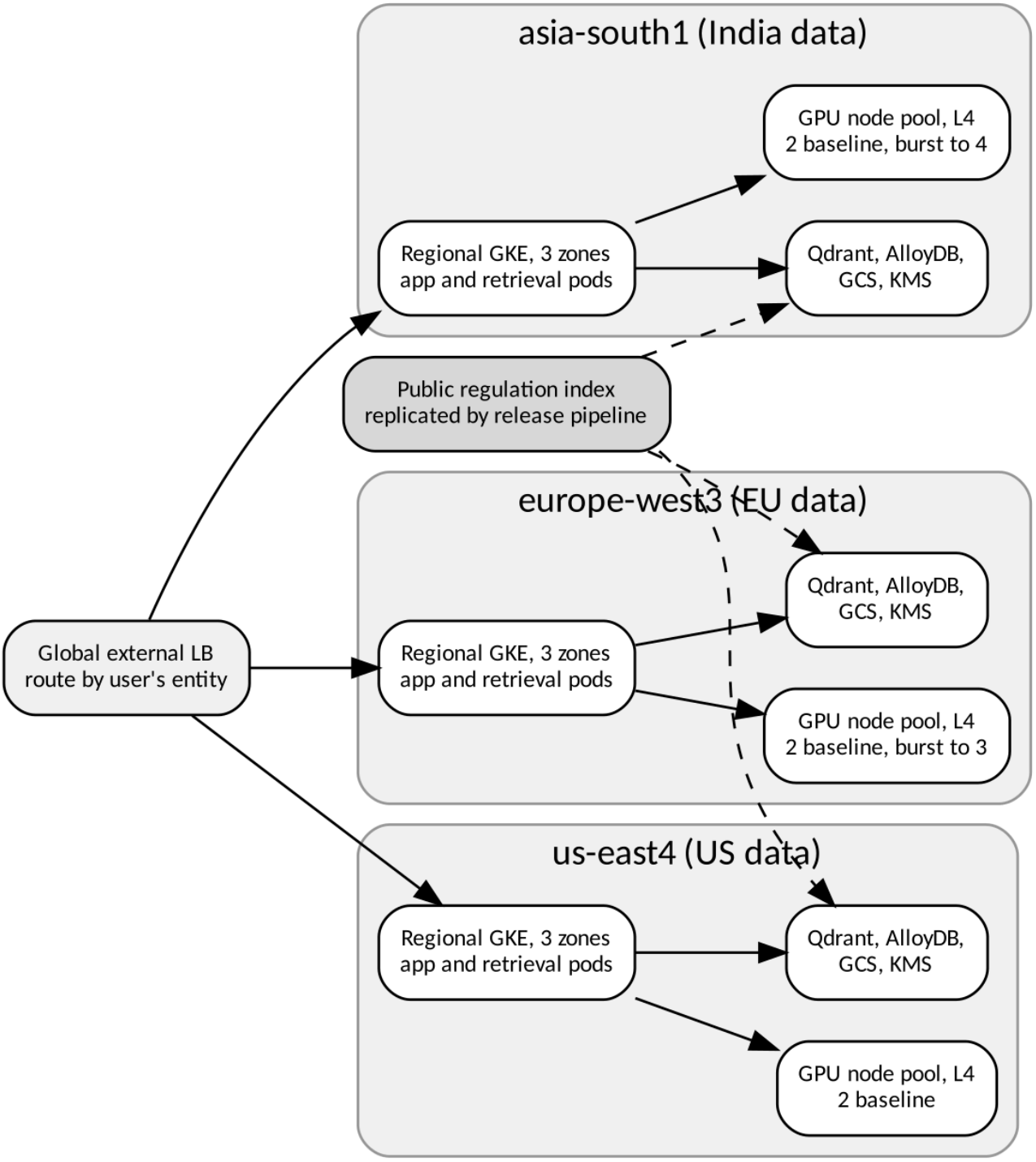
Records are kept for the longest period any rule requires. RBI KYC directions require transaction records for at least five years. FinServ's legal team should confirm the period for AI decision logs.

AI governance. A list of all models, a named owner for each, documented limits, regular bias and quality reviews, and a person making every final compliance decision. The system supports decisions; accountability stays with the compliance officer.

8. Infrastructure

8.1 Cloud and deployment

GCP, for GKE's mature GPU support, its regions in Mumbai, Frankfurt and Northern Virginia, and the option of regional Vertex AI services. Open-source components keep the application portable.



Layer	GCP service	Open-source alternative
Orchestration	GKE Standard, regional	Any Kubernetes
Inference	vLLM on G2 (NVIDIA L4)	Same

Layer	GCP service	Open-source alternative
Vector search	Qdrant on GKE	Same
Database	AlloyDB (**proprietary**)	PostgreSQL
Pipelines	Cloud Composer (**managed Airflow**)	Apache Airflow
Keys	Cloud KMS (**proprietary**)	HashiCorp Vault
Edge security	Cloud Armor (**proprietary**)	Coraza WAF
OCR	Document AI (**proprietary, optional**)	Tesseract, Docling

Each region runs on its own. An outage in India cannot fail over to the EU, because that would move Indian data abroad. High availability therefore comes from three zones inside each region and a defined degraded mode. This is why each region has its own baseline GPUs in the cost model.

8.2 Scaling the models

Component	Signal	Behaviour
vLLM pods	More than 3 requests waiting, or GPU cache above 80%	Scale up within 60 s; scale down after 10 quiet minutes
GPU nodes	Pods waiting for a node	Cluster Autoscaler; model image pre-loaded on disk to start faster
API and RAG pods	CPU and concurrent requests	Horizontal Pod Autoscaler
Embedding and re-rank	Queue length	Own node pool, because it scales with query count, not answer length
Nightly jobs	Schedule	Evaluation and re-embedding run outside business hours

GPU utilisation is a poor scaling signal for vLLM, because a running engine holds GPU memory and always looks busy. Queue length shows how long users wait. Each region keeps at least two replicas, because loading a model takes minutes. Under overload the system first switches to the smaller model, then to extractive answers.

8.3 Capacity and cost (500 users, 10,000 queries a day)

- **Average load:** 10,000 queries over a 10-hour day = 0.28 requests a second.
- **Peak load:** 500 concurrent users with 75 seconds between requests = 6.7 requests a second.
- **Peak generation:** 6.7 × 300 output tokens = about 2,000 tokens a second, across three regions.
- **Assumed GPU capacity:** 700 to 1,200 output tokens a second per L4 for a quantised 30B-class model. **This is an assumption and must be confirmed by a load test.**

The gap between 0.28 and 6.7 means the load comes in bursts. That supports scaling on queue length from a small baseline.

Item	USD a month	Basis
GPU inference, baseline (6 L4, 1-year commitment)	3,818	sourced
GPU inference, burst (3 L4, 220 h on demand)	687	sourced
GKE regional cluster fee	219	sourced
Application and retrieval nodes	2,956	estimate

Item	USD a month	Basis
Qdrant nodes	1,971	estimate
AlloyDB, 4 vCPU-equivalent with HA	2,836	derived
Storage, logging, KMS, network, archive	1,500	estimate
Total	**13,989**	

That is about **$0.064 a query** across 220,000 queries in a 22-business-day month. GPUs are 32% of the total.

How firm the numbers are. Only the GPU, GKE and AlloyDB lines are based on published prices, and the AlloyDB price for Mumbai comes from a third-party tracker. Other regions and the rows marked "estimate" need the GCP pricing calculator. Staff, support plans and taxes are not included.

Self-hosting does not save money at this volume. At 1,500 input and 300 output tokens per query, a regional managed model would cost about $290 to $1,597 a month in tokens, compared with about $4,506 for self-hosted GPUs. The extra cost buys residency control, fixed model versions and independence from a provider's retirement schedule. If legal approves managed inference for public text, a hybrid setup could cost less, and the gateway makes that a configuration change. This is open decision 1 in Section 12.

Keeping costs down: cache answers to repeated questions on public regulation (keyed by version), limit prompt size, send simple queries to the fast tier, scale down outside business hours, and report spend per team.

8.4 Monitoring

Area	Tools	What is measured	Alert
Infrastructure	Prometheus, Grafana	GPU memory, queue length, pod restarts	Queue over 10 for 2 minutes
LLM performance	OpenTelemetry, vLLM metrics	Time to first token, tokens a second, p95 latency	SLO budget burning too fast
Cost	Gateway token counter	Tokens and cost per team and per answer	80% of budget used
Traces	Langfuse (open source)	Prompt version, chunks, model, output	Sampled for review
Quality	Evaluation service	Refusal rate, validation failures, degraded answers	Refusal or failure rate doubles

Traces are redacted like logs and stay in the region. LangSmith and Arize are proprietary options if hosted tools are approved.

8.5 Evaluation pipeline

When	Trigger	Gate
Before merge	Change to prompt, config or code	Test set and regression tests must pass
Before release	New model or embedding	Shadow run, then staged rollout with automatic rollback
After ingestion	New regulation	New test questions added; earlier answers re-checked
In production	Daily sample of traffic	RAGAS-style scoring; low scores go to reviewers
Quarterly	Audit	Compliance reviews a sample of decisions

An **ablation test** runs in the pipeline, so every retrieval step must keep proving it is worth its cost. It has already helped: in the prototype it showed that the offline feature-based re-ranker did not improve on RRF alone (see the Evaluation Report). Without that comparison, the re-ranker would have looked fine.

8.6 Detecting retrieval drift

Drift	Signal	Response
Query drift	Population Stability Index on query embeddings; new topic clusters	Add test questions; review coverage
Corpus drift	Share of results from outdated documents	Check ingestion and filters
Relevance drift	Lower top scores, more refusals, lower recall on a fixed check set	Investigate; consider re-indexing
Feedback drift	Officer thumbs-down, citation corrections	Review queue; add to the test set
Embedding drift	Fixed test queries re-run nightly	Block unannounced model changes

8.7 Failure handling

Failure	Detected by	Response	What users see
GPU node lost	Health probes	Reschedule; the other replica serves	Slower answers
All LLM replicas down	Circuit breaker	Extractive mode	Quoted sources, marked degraded
Qdrant node lost	Cluster health	Replica serves; rebuild from snapshot	No change
Wrong regulation released	Regression tests, reviewer	Reopen the previous version using effective dates	Corrected after rollback
Bad prompt or model	Rollout gates	Automatic rollback	Only the rollout share is affected
Whole region lost	Multi-zone failure	Declare an incident; no cross-border failover	Reduced service in that region

9. Delivery roadmap

Phase	Scope	Done when
0. Discovery (3 weeks)	Workshops, legal and residency sign-off, test set written with officers	Scope and quality gates agreed
1. Pilot (10 weeks)	India, RBI corpus, Q&A with citations, 10 officers	Gates pass on officer-written questions
2. Screening (10 weeks)	Rules with Compliance, EU region, MiFID II	Rules agree with officers at the target level
3. Scale (10 weeks)	US region, reports, change impact, full SSO	Service levels met under load test
4. Operate	Continuous evaluation, model reviews	Quarterly governance review

Success is measured against the 60% search-time baseline, citation accuracy in officer reviews, and audit findings. Time saved must be measured with officers, not assumed.

10. What the prototype includes

Capability	Status
Versioned ingestion, paragraph chunking, as-of queries	Built and tested
Qdrant, BM25, RRF, re-ranking, compression, refusal	Built and tested
Citations, PII removal, injection screen, citation and number checks	Built and tested
Model router with residency, budget, circuit breaker and fallback	Built and tested, including against a stub HTTP server
LangGraph agent with fixed rules and follow-up questions	Built and tested
Regulatory change impact and audit-committee report	Built and tested
Hash-chained audit log	Built and tested
Ollama and vLLM client	Tested against a stub; runs on a real model with `make full`
Official PDF parsing	Tested on a generated PDF; `make full` handles live documents
Cross-encoder re-ranker	Built; needs a model download and runs with `make full`
GKE, vLLM scaling, VPC controls, customer-managed keys, Composer	Design only; not deployed
Permission filtering, entailment hallucination check	Design only
Multi-region deployment, load test results	Design only; capacity figures are assumptions

Limits of the regulatory content. The bundled corpus holds digests of official sources, labelled as such, with paragraph references. They are not the exact regulatory text and must not be used for real compliance decisions. Wire-transfer amount thresholds and CTR limits are left out, because they could not be confirmed from the available source text. The screening rules are examples and need validation by FinServ Compliance. High-risk country lists change after each FATF plenary, so production needs a controlled feed for them.

11. Risks

Risk	Likelihood	Impact	Mitigation
Wrong regulation cited	Medium	High	Paragraph citations, number checks, human review
Officers trust the tool too much	Medium	High	Show sources first, track overrides, keep officers accountable
Rules go out of date	Medium	High	Change impact links rules to paragraphs and alerts owners
Cloud outsourcing conditions	Medium	High	Legal review in Phase 0; portable open-source stack
GPU supply or cost rises	Medium	Medium	Provider-neutral gateway, several model tiers
Open model licence changes	Low	Medium	Pin model weights; keep a second approved model

12. Open decisions for FinServ

1. Can public regulatory text use regional managed inference, given the cost difference in 8.3?
2. Who owns the rule set and signs it off?

3. Which US regulations are in scope?
4. How long must AI decision logs be kept?
5. What service is acceptable in degraded mode during a regional outage?

13. Architecture Decision Records

ADR-001: Vector database selection

Status: Accepted, subject to a load test in Phase 1. **Date:** November 2026.

Context. The vector store must:

- keep each region's data in that region;
- filter on date, framework and user permissions on every query, because point-in-time questions are central;
- support both dense and keyword search;
- handle an initial corpus of a few million chunks.

The team has limited Kubernetes capacity, so the effort to run it matters.

Decision. Use self-hosted **Qdrant** on GKE in each region, with three replicas. In production, dense and sparse search run in one query. The prototype uses Qdrant's embedded mode, with BM25 kept separately.

Alternatives considered

Option	For	Against
Qdrant	Strong filtering, built-in sparse vectors, same engine on a laptop and in a cluster, Apache 2.0	One more stateful system to run
pgvector on AlloyDB	One database, joins with other tables, familiar	Filtered search needs careful tuning at scale; weaker built-in hybrid search
Weaviate	Strong built-in hybrid search	Heavier to run at this size
Milvus	Built for very large scale	Too many parts for a few million chunks
Pinecone	Fully managed	Proprietary; no self-hosted option; data location depends on the vendor
Vertex AI Vector Search	Managed in GCP	Proprietary, less portable, not allowed in the open-source prototype

Consequences

- Good: one engine handles dense search, sparse search and date filters, with the same API from laptop to production.
- Good: each region gets its own deployment, with snapshots kept in that region.
- Cost: FinServ is responsible for upgrades, backup and recovery.
- Cost: relational data stays in AlloyDB, so a registry update and a vector update are not one transaction. A repeatable release job keeps them in step.
- Revisit if the team cannot support Qdrant, or if a pgvector benchmark on real filters performs as well, which would remove one component.

ADR-002: Build versus buy for the orchestration layer

Status: Accepted. **Date:** November 2026.

Context. The orchestration layer connects rules, retrieval, models and the audit log. Regulators expect decisions to be repeatable. The client has warned against repackaged framework tutorials and against depending on one LLM provider. The options range from a managed agent platform, to a framework, to fully custom code.

Decision. A mix of both:

- **Use** open-source **LangGraph** only to run the graph, hold state and save checkpoints.
- **Build** the parts that carry the compliance value: rule engine, retriever, LLM gateway, guardrails and audit log. These have plain Python interfaces, and no framework types cross them.

Alternatives considered

Option	For	Against
LangGraph plus own services	Clear graph, replayable trace, few dependencies	Team must learn the graph model; library upgrades need care
LangChain agents throughout	Quick start, many integrations	Hides the logic an auditor needs to understand
CrewAI	Quick multi-agent setup	Conversation-based flow is hard to make repeatable
Haystack	Strong RAG pipelines	Less suited to stateful flows with human checkpoints
Vertex AI Agent Builder	Managed, fast to deploy	Proprietary; ties orchestration to one vendor
Fully custom code	No dependency	Must rebuild checkpoints, retries and tracing

Consequences

- Good: LangGraph can be replaced without rewriting compliance logic, because each step calls an ordinary service.
- Good: the path is fixed and every step is traced, so any decision can be replayed.
- Cost: the agent is less flexible than a free-running one. A new workflow needs a graph change and a review. This is accepted on purpose.
- Cost: the services we build need maintenance and tests. The prototype's test suite is the start of that.
- Revisit if managed platforms offer runs that are region-pinned, exportable and replayable enough to satisfy audit.

ADR-003: Guardrail implementation approach

Status: Accepted. **Date:** November 2026.

Context. The worst failure for this system is an invented number or a wrong citation in a compliance answer. Regulators expect evidence, so each guardrail decision must itself be explainable. Guardrails run on every request, so they must be fast. Checks done entirely by an LLM add delay, cost, and a second model that can also be wrong.

Decision. Build **layered guardrails into the application, with rule-based checks first**:

- **Always run:** PII patterns, injection patterns, citation validity, numbers present in the cited source, and schema validation.
- **On failure:** retry once, naming the failed check; then fall back to quoted source text.
- **In production, add** a model-based entailment check as a second layer. It can reject an answer, but it cannot approve one that failed a rule-based check.

Alternatives considered

Option	For	Against
Rules first, model second	Fast, explainable, testable, no extra GPU	Rules miss subtle errors, so a second layer is still needed
NeMo Guardrails	Rich flow control and its own rules language	Adds a framework and a new language for a small set of checks
Guardrails AI	Good validator library, strong on schemas	Regulatory checks would still be custom
LLM-as-judge on every answer	Catches meaning errors	Adds cost and delay; results vary between runs
Provider safety filters only	No build effort	Not aware of regulation; ties control to one provider
Review after the fact only	Simple	Errors reach users first

Consequences

- Good: every rejection has an exact reason that can go in the audit log.
- Good: the guardrails were tested without any LLM, including a stub answer with an invented number, which was rejected.
- Cost: number matching can reject a correct calculated value. The fallback then quotes the source, which is less helpful but safe.
- Cost: passing every rule-based check does not prove an answer is right. A wrongly paraphrased clause with valid citations could pass. That is why the entailment check and human review stay in the design.
- Revisit if the entailment check rejects too many correct answers, beyond the level agreed with Compliance.

14. References

Regulatory sources

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